

Nairobi Flood Intelligence v1.0

— Citywide Subscriber Experience

This release expands the product from a single-score demo into a citywide, per-area decision-support prototype covering 60 Nairobi neighbourhoods.

Product flow

Subscriber → saves places (matched to one of 60 monitored areas) → sees current risk for those areas → sees recommended action → monitors routes between areas → reports flooding.

Area coverage and data provenance

Zone susceptibility (baseline, drainage, historical exposure, river exposure, elevation) is a first-pass editorial judgment informed by:

- The 2026 Nairobi Rivers Regeneration Programme mapping of 38 flood-prone neighbourhoods across five river corridors (Nairobi East, West, North,

Central and South), reported by the Ministry of Interior.

- The 2024 Space4All "Diagnostic of Urban Flooding in Nairobi" report, which names Kibera, Mathare, Mukuru and Kariobangi as the most consistently affected communities and estimates roughly 27% of city territory (35% of built-up area) as highly flood-prone.
- Public reporting on the March–May 2024 and March–May 2026 Nairobi flood events (Nairobi and Ngong rivers overflowing, major roads and JKIA disrupted).

22 additional Nairobi areas outside the official mapping are included for citywide coverage, generally scored lower where elevation and drainage are better documented as favourable (e.g. Karen, Runda, Muthaiga).

These susceptibility scores are not calibrated hydrological output. They are a defensible starting point pending real terrain/drainage modelling — see "Next production work" below.

Current integrations

- Live Nairobi hourly + 15-minute rainfall/forecast via Open-Meteo, applied against all 60 zones.

- Historical Nairobi flood observations via the public ArcGIS FeatureServer.
- Local browser storage for saved places, routes, and prototype flood reports (geolocated and assigned to the nearest monitored zone).

Design principles

1. Alerts must be relevant to a subscriber's location/route.
2. Every alert should explain where, why, when and what action to take.
3. Risk score is an index, not a probability.
4. Official warnings remain authoritative.
5. Community reports must be verified before becoming high-confidence alerts.

Next production work

- Account/authentication and paid subscriptions.
- SMS/WhatsApp delivery.
- Geocoded saved places (currently matched to the nearest of 60 named zones, not a real address) and actual route geometry (currently a straight line between two zones).
- Calibrated, per-zone susceptibility from real terrain/drainage modelling, replacing the current

editorial first-pass scores.

- Per-zone rainfall rather than one citywide reading applied to all areas.
- KMD official-warning ingestion (currently always reads "no live feed").
- Timestamped historical flood events (the ArcGIS layer currently has no per-point timestamp).
- Human/automated report verification (all community reports are currently unverified by design).
- Alert history and subscriber feedback ("useful / not useful").
- Fleet dashboard and API.